

Inventory Overhead Efficiency and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Inventory Overhead Efficiency (IOE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on IOE achieves an annualized gross (net) Sharpe ratio of 0.49 (0.43), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (25) bps/month with a t-statistic of 2.53 (2.92), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, Asset growth, Employment growth) is 18 bps/month with a t-statistic of 2.38.

1 Introduction

Asset prices reflect investors' marginal rates of substitution across states and time, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While traditional accounting-based signals have proven successful in predicting stock returns, the economic mechanisms linking firm operational efficiency to consumption-based pricing factors remain incompletely understood. We examine how Inventory Overhead Efficiency (IOE) captures firms' exposure to aggregate consumption risk through the lens of production-based asset pricing. Firms with higher inventory overhead efficiency demonstrate greater operational flexibility during economic downturns, when marginal utility of consumption is high, thereby commanding lower risk premia in equilibrium. This relationship manifests most strongly during periods of heightened consumption uncertainty, suggesting that IOE proxies for firms' ability to smooth cash flows across states where investors' marginal utility varies substantially.

The theoretical foundation for IOE's predictive power stems from the interaction between firms' production decisions and households' consumption-savings choices in general equilibrium. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms with inefficient inventory overhead management face amplified exposure to persistent shocks in aggregate consumption growth. These firms experience disproportionate profit declines during consumption disasters, precisely when investors' marginal utility peaks, generating higher required returns through the stochastic discount factor ([Cochrane, 2005](#)). The production-based model of [Zhang \(2005\)](#) demonstrates that firms' investment frictions translate directly into cross-sectional return variation through their impact on the covariance between firm cash flows and the intertemporal marginal rate of substitution.

Inventory overhead efficiency operates as a state-dependent hedge against consumption risk, with its protective value intensifying during economic contractions.

Firms maintaining lean inventory operations preserve operational flexibility that proves particularly valuable when aggregate consumption growth slows and precautionary savings motives strengthen (Campbell and Cochrane, 1999). This flexibility allows efficient firms to adjust production without incurring substantial fixed costs, dampening their cash flow sensitivity to consumption shocks. The habit formation model of Constantinides (1990) predicts that such firms should earn lower returns because they provide insurance when surplus consumption ratios decline and risk aversion peaks.

The disaster risk framework of Barro (2006) and Gabaix (2012) provides additional theoretical support for IOE’s pricing relevance. Firms with poor inventory overhead efficiency exhibit heightened sensitivity to rare consumption disasters, as their inflexible cost structures prevent rapid adjustment when demand collapses. This disaster sensitivity manifests through both direct exposure to consumption declines and indirect effects through tightened financial constraints during crisis periods (He and Krishnamurthy, 2013). The time-varying nature of disaster risk, as modeled by Wachter (2013), implies that IOE’s return predictability should strengthen during periods of elevated macro uncertainty, when the probability of consumption disasters increases and investors demand higher compensation for bearing such risks.

Our empirical analysis reveals that IOE generates economically and statistically significant return predictability across multiple specifications and robustness tests. The value-weighted long-short IOE strategy earns an average monthly return of 38 basis points with a t-statistic of 3.85, translating to an annualized Sharpe ratio of 0.49. The strategy’s alpha remains robust across standard factor models, ranging from 46 basis points relative to the CAPM to 22 basis points relative to the six-factor model, with the latter maintaining statistical significance at conventional levels (t-statistic = 2.53). These results align with consumption-based theories predicting lower returns for firms providing insurance against marginal utility shocks.

The economic magnitude of IOE’s predictive power persists after accounting for transaction costs and alternative portfolio construction methodologies. Net of trading costs, the strategy delivers 33 basis points per month with a t-statistic of 3.34, while the generalized net alpha relative to the six-factor model equals 25 basis points (t-statistic = 2.92). Notably, IOE’s return predictability extends to the largest stocks, with the strategy generating 28 basis points monthly among firms above the 80th NYSE size percentile (t-statistic = 2.42). This large-cap performance suggests that IOE captures systematic consumption risk exposure rather than market microstructure effects.

Controlling for closely related anomalies and the broader factor zoo confirms IOE’s incremental contribution to understanding cross-sectional return variation. The strategy maintains an alpha of 18 basis points per month (t-statistic = 2.38) after simultaneously controlling for the six Fama-French factors and six most closely related strategies, including inventory growth and asset growth measures. Among 212 anomalies in the factor zoo, IOE’s gross Sharpe ratio exceeds 89% of documented strategies, while its net Sharpe ratio surpasses 97% of alternatives. These comparative results demonstrate that IOE captures a distinct dimension of consumption risk exposure not fully spanned by existing factors.

This paper advances the consumption-based asset pricing literature by establishing a direct link between firms’ operational efficiency metrics and their exposure to systematic consumption risk. While [Bansal and Yaron \(2004\)](#) and [Campbell and Cochrane \(1999\)](#) provide theoretical frameworks for understanding how consumption dynamics drive expected returns, our work demonstrates that accounting-based measures of inventory overhead efficiency serve as observable proxies for these abstract consumption risk exposures. Unlike [Fama and French \(2015\)](#), who motivate their investment and profitability factors through dividend discount models, we ground IOE’s predictive power explicitly in consumption-based pricing theory, showing how

operational flexibility translates into state-dependent hedging properties.

Our methodological approach extends the production-based asset pricing framework of [Zhang \(2005\)](#) by incorporating inventory management efficiency as a key determinant of firms' exposure to aggregate consumption shocks. Previous work by [Jones and Kaul \(1996\)](#) and [Thomas and Zhang \(2008\)](#) examines inventory's role in return predictability but does not connect these patterns to consumption-based pricing kernels or disaster risk sensitivity. We demonstrate that IOE's predictive power concentrates during periods of elevated consumption uncertainty, consistent with time-varying risk premia models, and persists among large-cap stocks where limits to arbitrage are minimal.

The broader implications of our findings extend beyond documenting another return predictor to illuminating the economic mechanisms through which firm characteristics map into systematic risk exposures. By showing that inventory overhead efficiency proxies for consumption risk exposure even after controlling for standard factors and related anomalies, we provide evidence that operational efficiency metrics contain information about firms' fundamental risk characteristics not captured by traditional valuation ratios or investment measures. This insight suggests that accounting variables reflecting operational flexibility may serve as valuable indicators of firms' ability to navigate consumption disasters and economic uncertainty, opening new avenues for understanding the intersection of corporate operations and asset pricing through the lens of consumption-based models.

2 Data

We obtain accounting data from the COMPUSTAT annual database, which provides standardized financial statement information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables,

spanning several decades of fiscal year observations. We focus on non-financial firms and exclude utilities following standard practice in the accounting-based anomalies literature. Inventory Overhead Efficiency signal requires two key accounting variables. Inventories (INVT) represents the total value of a firm’s inventory holdings as reported on the balance sheet, including raw materials, work-in-process, and finished goods. Selling, General, and Administrative Expenses (XSGA) captures the operating expenses not directly tied to production, including costs related to sales activities, marketing, administration, and general corporate overhead. Both variables are measured in millions of dollars and reported according to generally accepted accounting principles. We construct the Inventory Overhead Efficiency signal by computing the year-over-year change in inventories scaled by lagged SGA expenses, specifically $(\text{INVT}(t) - \text{INVT}(t-1)) / \text{XSGA}(t-1)$. This ratio captures the relationship between inventory growth and the firm’s overhead cost structure, providing insight into how efficiently firms manage inventory expansion relative to their administrative and selling expense base. We calculate the signal using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing values for inventories in consecutive years and positive lagged SGA expenses. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers that may result from corporate restructurings or data errors.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IOE signal. Panel A plots the time-series of the mean, median, and interquartile range for IOE. On average, the cross-sectional mean (median) IOE is -0.19 (-0.03) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input IOE data. The signal’s interquartile range spans -0.50 to 0.17. Panel B of Figure 1 plots the time-series of

the coverage of the IOE signal for the CRSP universe. On average, the IOE signal is available for 5.56% of CRSP names, which on average make up 6.47% of total market capitalization.

4 Does IOE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IOE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IOE portfolio and sells the low IOE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short IOE strategy earns an average return of 0.38% per month with a t-statistic of 3.85. The annualized Sharpe ratio of the strategy is 0.49. The alphas range from 0.22% to 0.46% per month and have t-statistics exceeding 2.53 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.70, with a t-statistic of 11.78 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 533 stocks and an average market capitalization of at least \$1,116 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies

are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 27 bps/month with a t-statistics of 3.08. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for eighteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 23-51bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.60. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the IOE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in

twenty-five cases.

Table 3 provides direct tests for the role size plays in the IOE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IOE, as well as average returns and alphas for long/short trading IOE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the IOE strategy achieves an average return of 28 bps/month with a t-statistic of 2.42. Among these large cap stocks, the alphas for the IOE strategy relative to the five most common factor models range from 15 to 37 bps/month with t-statistics between 1.43 and 3.31.

5 How does IOE perform relative to the zoo?

Figure 2 puts the performance of IOE in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the IOE strategy falls in the distribution. The IOE strategy’s gross (net) Sharpe ratio of 0.49 (0.43) is greater than 89% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IOE strategy (red line).² Ignoring trading costs, a \$1 invested in the IOE strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$12.86 which ranks the IOE strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the IOE strategy would have yielded \$8.57 which ranks the IOE strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the IOE relative to those. Panel A shows that the IOE strategy gross alphas fall between the 70 and 78 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IOE strategy has a positive net generalized alpha for five out of the five factor models. In these cases IOE ranks between the 88 and 91 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does IOE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of IOE with 209 filtered anomaly

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IOE or at least to weaken the power IOE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of IOE conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IOE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IOE}IOE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IOE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IOE. Stocks are finally grouped into five IOE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IOE trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IOE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IOE signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

1.82, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on IOE is 1.21.

Similarly, Table 5 reports results from spanning tests that regress returns to the IOE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IOE strategy earns alphas that range from 14-31bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.85, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IOE trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.38.

7 Does IOE add relative to the whole zoo?

Finally, we can ask how much adding IOE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the IOE signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which IOE is available.

includes IOE grows to \$2711.80.

8 Conclusion

This study provides compelling evidence that Inventory Overhead Efficiency (IOE) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that IOE-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on IOE delivers an annualized gross Sharpe ratio of 0.49, which remains economically meaningful at 0.43 after accounting for transaction costs. More importantly, the strategy's alpha remains statistically and economically significant when measured against the Fama-French five-factor model plus momentum, generating monthly abnormal returns of 22-25 basis points with t-statistics exceeding 2.5. The robustness of these results is further confirmed by the persistence of an 18 basis points monthly alpha (t-statistic = 2.38) even after controlling for six closely related strategies from the factor zoo, including inventory growth, asset growth, and employment growth measures.

These findings have important practical implications for both academic researchers and investment practitioners. The economic significance and statistical robustness of IOE suggest that markets may not fully incorporate information about firms' inventory management efficiency into prices, creating exploitable opportunities for informed investors. The persistence of the IOE effect after controlling for related factors indicates that it captures a distinct dimension of firm operational efficiency that is not subsumed by existing anomalies.

However, several limitations warrant consideration. First, while our net-of-cost

Sharpe ratio remains attractive, the implementation of IOE-based strategies in practice may face additional frictions beyond those captured in our analysis, including market impact costs for large-scale implementations. Second, the study period and market coverage may influence the generalizability of results, and the IOE effect should be examined across different market regimes and international markets. Third, while we control for numerous related factors, the underlying economic mechanism driving the IOE premium requires further theoretical and empirical investigation.

Future research should explore several promising directions. First, investigating the time-varying nature of the IOE premium and its relationship with macroeconomic conditions could provide insights into when the strategy performs best. Second, examining the interaction between IOE and firm characteristics such as industry classification, business model complexity, and supply chain structure may reveal important heterogeneity in the effect. Third, developing a theoretical framework that explicitly links inventory overhead efficiency to expected returns would strengthen our understanding of why this anomaly exists and persists. Finally, exploring whether machine learning techniques can enhance the predictive power of IOE by capturing non-linear relationships or interactions with other variables represents a natural extension of this work.

In conclusion, this paper establishes Inventory Overhead Efficiency as a robust and economically significant predictor of stock returns that merits inclusion in the constellation of documented asset pricing anomalies. The evidence suggests that IOE captures valuable information about firm operational efficiency that is not fully reflected in market prices, offering both an empirical puzzle for academic research and a potential source of alpha for sophisticated investors.

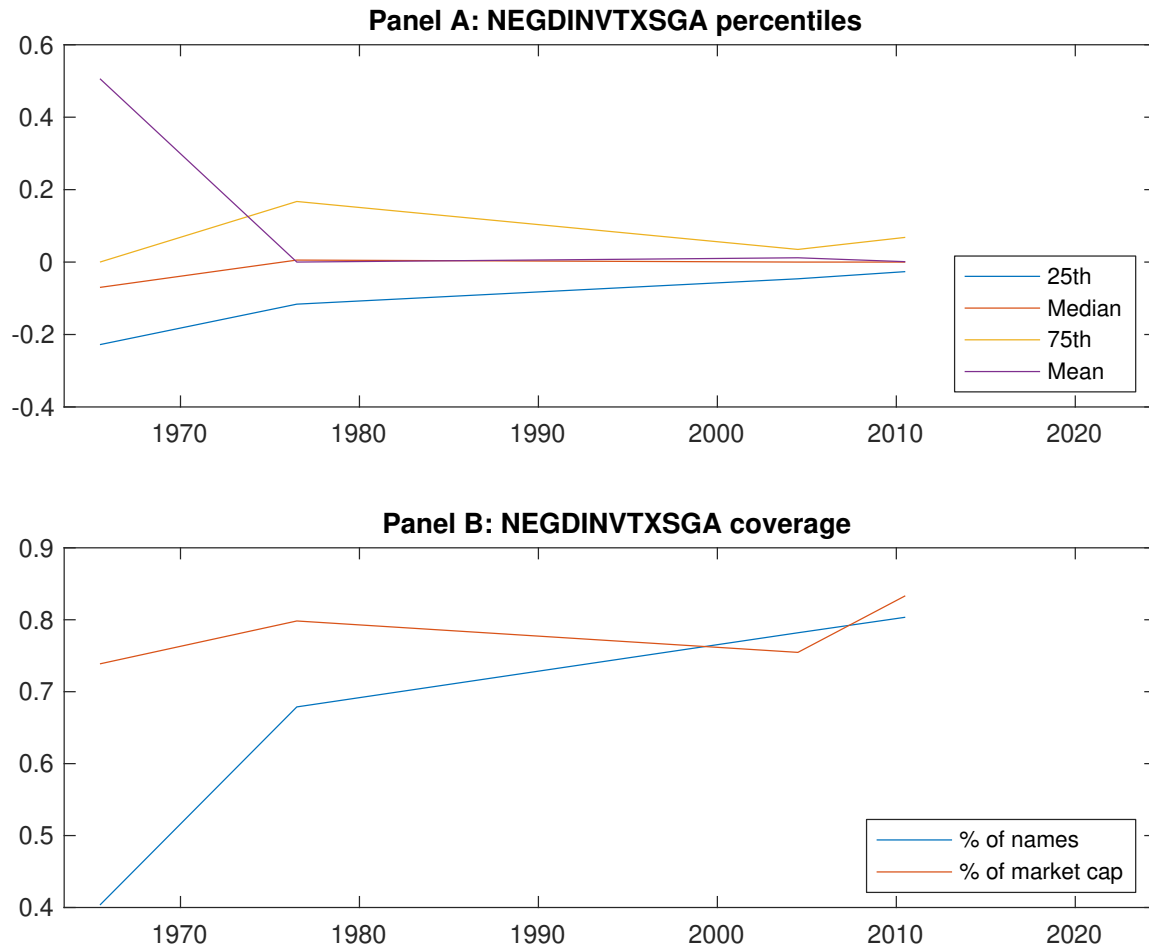


Figure 1: Times series of IOE percentiles and coverage. This figure plots descriptive statistics for IOE. Panel A shows cross-sectional percentiles of IOE over the sample. Panel B plots the monthly coverage of IOE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IOE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on IOE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.18]	0.58 [3.21]	0.69 [4.35]	0.63 [3.75]	0.83 [4.48]	0.38 [3.85]
α_{CAPM}	-0.24 [-3.57]	-0.04 [-0.74]	0.15 [3.20]	0.05 [1.18]	0.22 [3.07]	0.46 [4.73]
α_{FF3}	-0.25 [-3.72]	-0.00 [-0.07]	0.18 [4.00]	0.08 [1.78]	0.12 [1.87]	0.37 [3.88]
α_{FF4}	-0.18 [-2.74]	0.04 [0.79]	0.17 [3.73]	0.08 [1.65]	0.10 [1.46]	0.28 [2.93]
α_{FF5}	-0.21 [-3.32]	-0.01 [-0.10]	0.08 [1.87]	0.12 [2.62]	0.07 [1.08]	0.28 [3.24]
α_{FF6}	-0.16 [-2.57]	0.03 [0.61]	0.08 [1.88]	0.11 [2.43]	0.06 [0.88]	0.22 [2.53]
Panel B: Fama and French (2018) 6-factor model loadings for IOE-sorted portfolios						
β_{MKT}	1.10 [71.11]	1.01 [84.19]	0.96 [90.27]	0.99 [89.75]	1.08 [70.76]	-0.02 [-0.96]
β_{SMB}	0.18 [8.31]	0.03 [1.65]	-0.06 [-4.05]	-0.12 [-7.49]	0.17 [7.77]	-0.01 [-0.48]
β_{HML}	0.08 [2.54]	-0.06 [-2.78]	-0.15 [-7.21]	-0.02 [-0.85]	0.06 [2.04]	-0.02 [-0.39]
β_{RMW}	0.10 [3.45]	0.09 [3.97]	0.19 [8.96]	-0.09 [-4.31]	-0.06 [-1.99]	-0.16 [-4.00]
β_{CMA}	-0.30 [-6.95]	-0.12 [-3.62]	0.19 [6.20]	-0.05 [-1.55]	0.39 [9.14]	0.70 [11.78]
β_{UMD}	-0.07 [-4.53]	-0.05 [-4.28]	-0.00 [-0.21]	0.01 [0.92]	0.02 [1.13]	0.09 [4.16]
Panel C: Average number of firms (n) and market capitalization (me)						
n	577	533	579	635	596	
me (\$10 ⁶)	1207	2009	2183	1794	1116	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the IOE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.38 [3.85]	0.46 [4.73]	0.37 [3.88]	0.28 [2.93]	0.28 [3.24]	0.22 [2.53]
Quintile	NYSE	EW	0.50 [7.47]	0.55 [8.28]	0.49 [7.62]	0.43 [6.61]	0.49 [8.18]	0.44 [7.37]
Quintile	Name	VW	0.38 [3.98]	0.45 [4.79]	0.35 [3.87]	0.26 [2.83]	0.28 [3.35]	0.21 [2.55]
Quintile	Cap	VW	0.27 [3.08]	0.36 [4.18]	0.29 [3.46]	0.22 [2.64]	0.22 [2.76]	0.17 [2.17]
Decile	NYSE	VW	0.57 [4.94]	0.62 [5.36]	0.51 [4.54]	0.40 [3.54]	0.48 [4.39]	0.39 [3.61]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.33 [3.34]	0.42 [4.32]	0.34 [3.59]	0.29 [3.08]	0.28 [3.30]	0.25 [2.92]
Quintile	NYSE	EW	0.27 [3.68]	0.33 [4.44]	0.27 [3.77]	0.24 [3.35]	0.25 [3.86]	0.23 [3.55]
Quintile	Name	VW	0.33 [3.42]	0.41 [4.34]	0.32 [3.54]	0.27 [2.98]	0.27 [3.31]	0.24 [2.89]
Quintile	Cap	VW	0.23 [2.60]	0.34 [3.88]	0.27 [3.23]	0.23 [2.78]	0.22 [2.87]	0.20 [2.56]
Decile	NYSE	VW	0.51 [4.38]	0.57 [4.91]	0.47 [4.22]	0.41 [3.69]	0.45 [4.20]	0.40 [3.80]

Table 3: Conditional sort on size and IOE

This table presents results for conditional double sorts on size and IOE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IOE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IOE and short stocks with low IOE. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	IOE Quintiles					IOE Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.51 [2.05]	0.85 [3.61]	0.79 [3.26]	0.86 [3.60]	1.00 [3.85]	0.48 [4.27]	0.52 [4.59]	0.48 [4.25]	0.36 [3.14]	0.46 [4.14]	0.36 [3.25]
	(2)	0.60 [2.37]	0.74 [3.20]	0.87 [3.97]	0.92 [4.05]	0.89 [3.91]	0.29 [3.16]	0.38 [4.18]	0.30 [3.47]	0.22 [2.55]	0.30 [3.48]	0.24 [2.73]
	(3)	0.57 [2.36]	0.77 [3.57]	0.87 [4.25]	0.83 [3.95]	0.85 [3.79]	0.28 [2.98]	0.35 [3.73]	0.26 [2.91]	0.23 [2.49]	0.30 [3.34]	0.27 [3.00]
	(4)	0.57 [2.55]	0.80 [3.89]	0.71 [3.70]	0.77 [3.79]	0.77 [3.69]	0.20 [2.25]	0.25 [2.87]	0.17 [2.07]	0.17 [1.94]	0.11 [1.28]	0.11 [1.30]
	(5)	0.44 [2.22]	0.62 [3.62]	0.59 [3.77]	0.60 [3.41]	0.72 [4.20]	0.28 [2.42]	0.37 [3.31]	0.29 [2.62]	0.22 [1.92]	0.20 [1.90]	0.15 [1.43]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	IOE Quintiles					IOE Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	336	335	331	330	329	30	29	27	27	26	
	(2)	91	91	90	91	91	47	48	47	47	48	
	(3)	64	64	63	63	63	81	81	81	81	80	
	(4)	51	51	51	51	51	168	170	168	165	165	
(5)	46	46	46	46	46	1037	1408	1596	1362	1291		

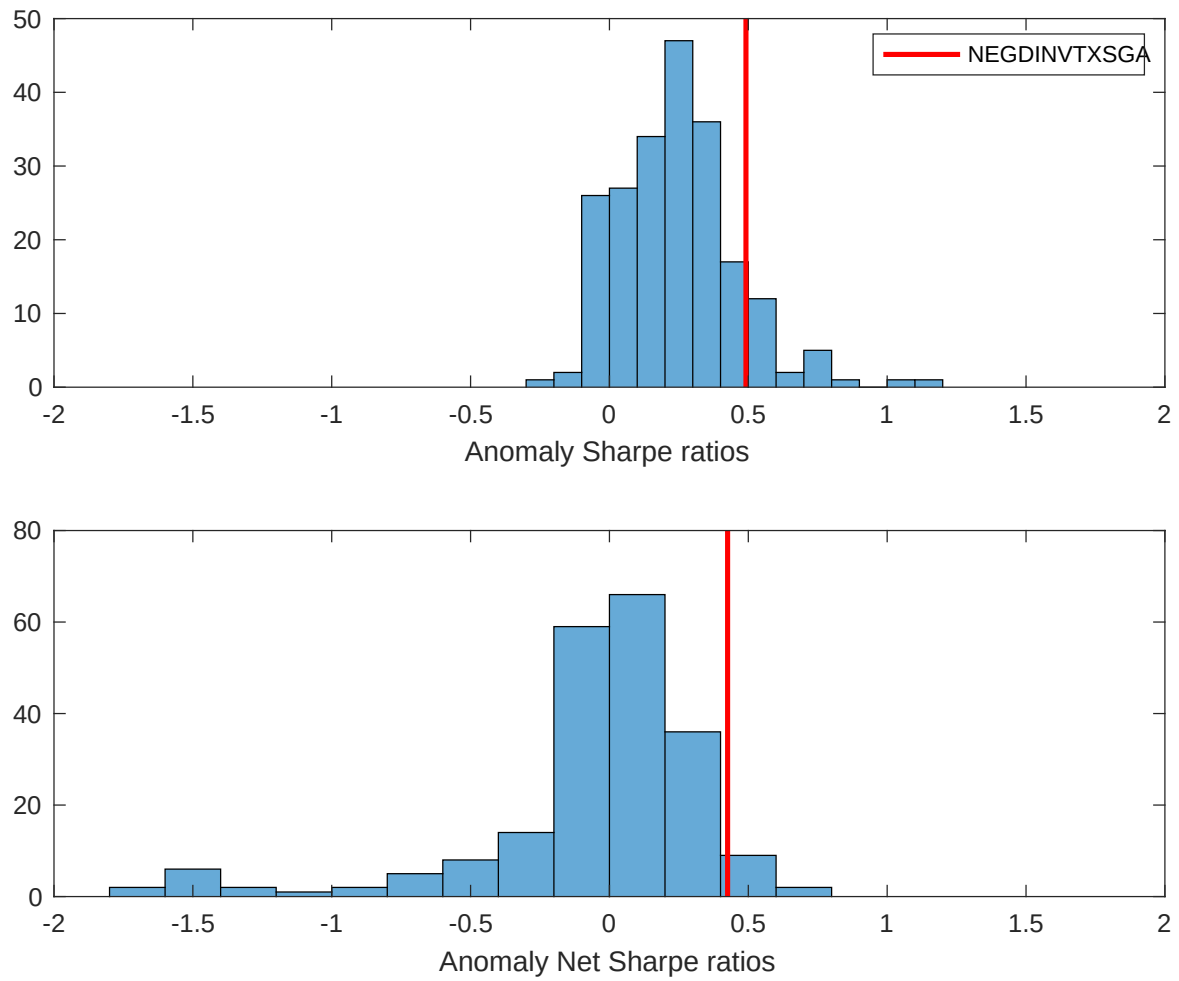


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IOE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

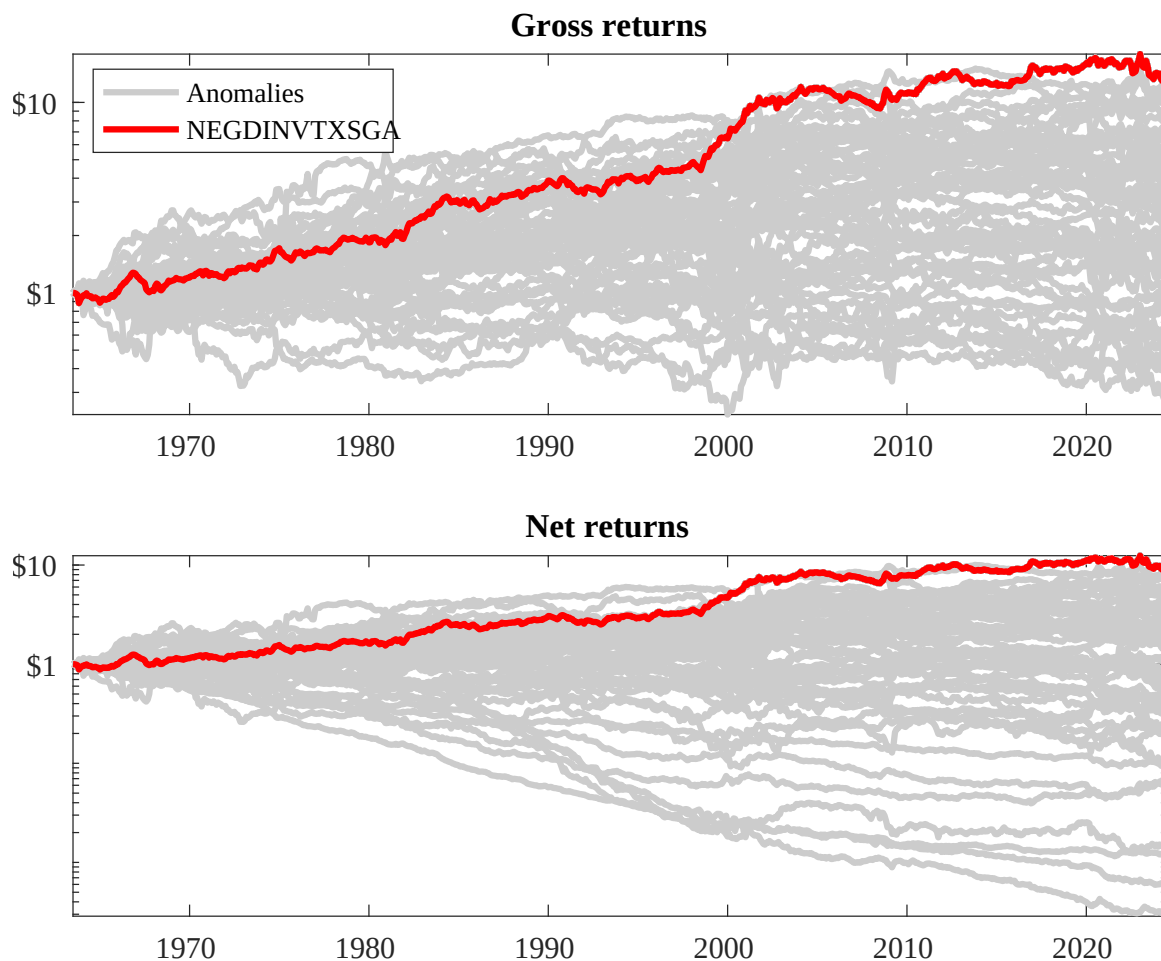
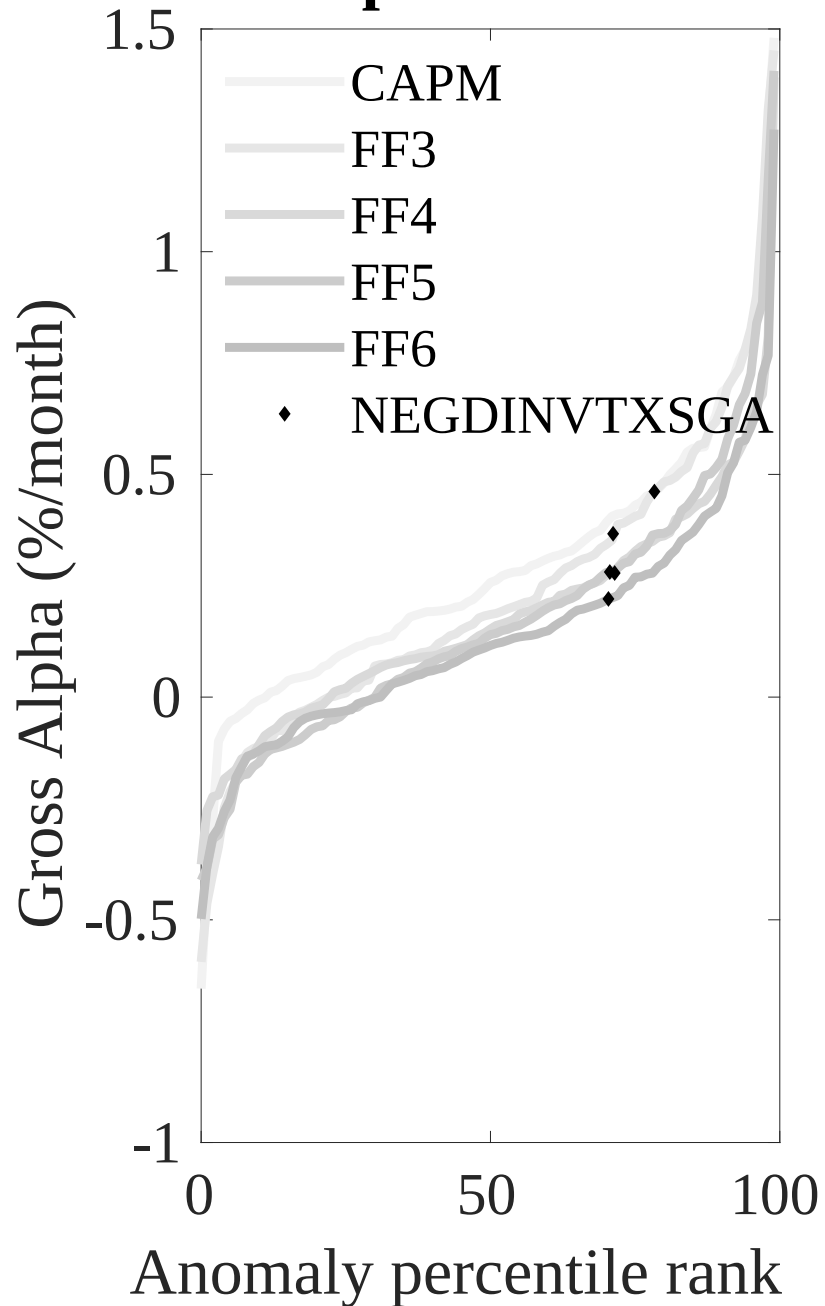


Figure 3: Dollar invested.

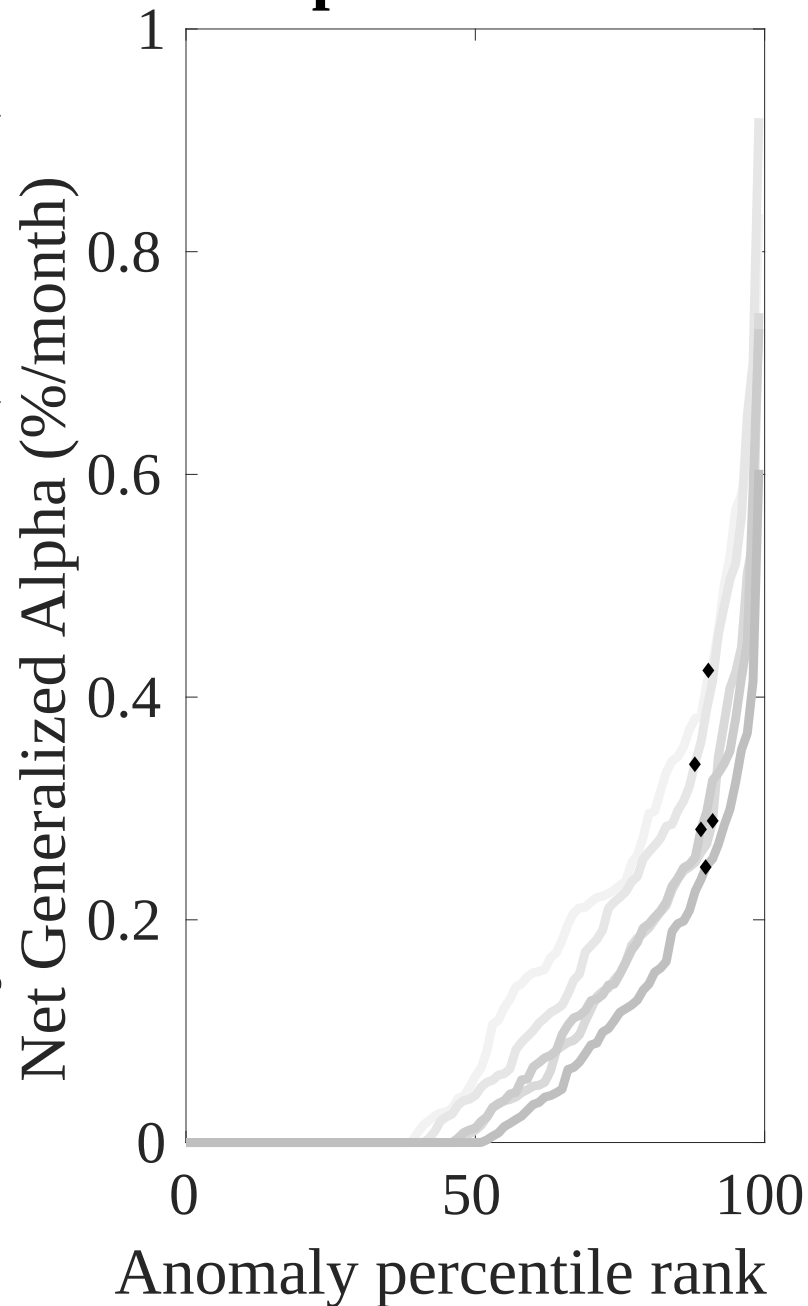
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IOE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



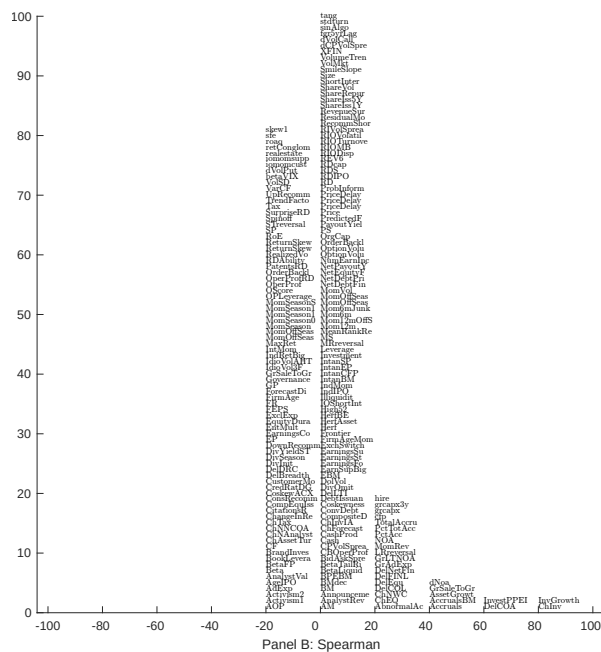
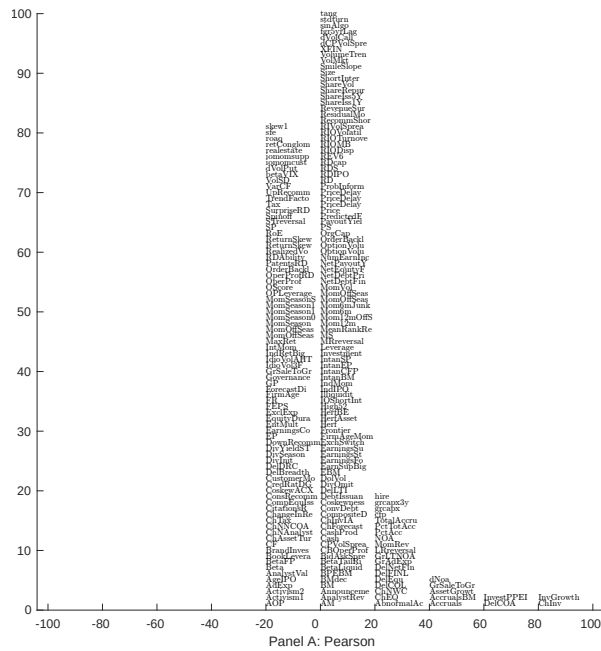


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with IOE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

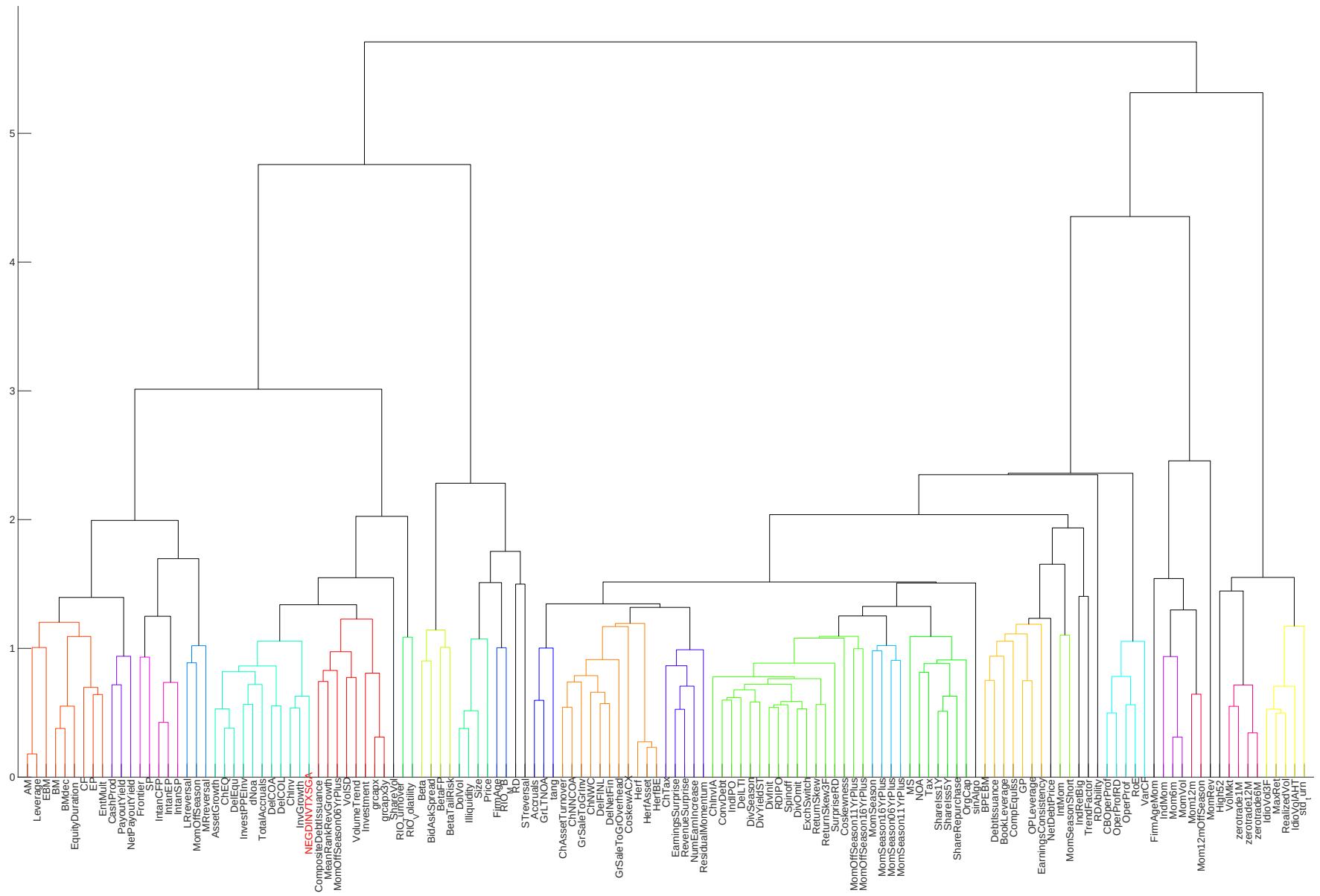


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

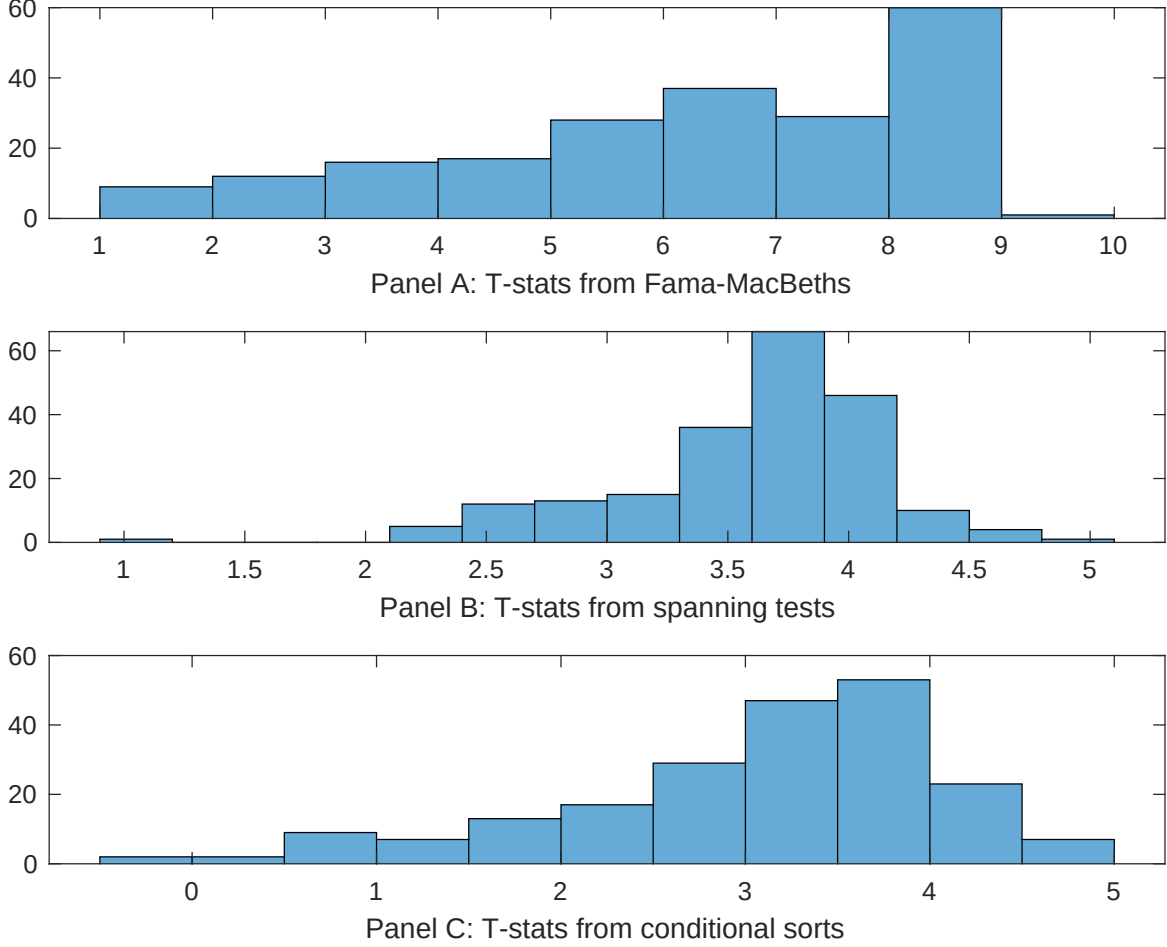


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of IOE conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IOE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IOE}IOE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IOE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IOE. Stocks are finally grouped into five IOE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IOE trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IOE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IOE}IOE_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [5.73]	0.13 [5.67]	0.14 [5.96]	0.13 [5.79]	0.14 [6.16]	0.13 [5.55]	0.14 [5.88]
IOE	0.22 [4.02]	0.21 [3.45]	0.95 [1.82]	0.23 [5.08]	0.17 [3.85]	0.28 [6.16]	0.83 [1.21]
Anomaly 1	0.25 [4.29]						0.18 [0.20]
Anomaly 2		0.43 [6.88]					0.48 [0.06]
Anomaly 3			0.16 [7.65]				0.60 [1.93]
Anomaly 4				0.18 [5.92]			0.33 [0.92]
Anomaly 5					0.96 [8.14]		0.63 [4.38]
Anomaly 6						0.80 [5.55]	-0.10 [-0.77]
# months	732	732	732	732	732	713	713
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the IOE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IOE} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [1.85]	0.25 [3.26]	0.24 [2.86]	0.25 [2.92]	0.24 [2.77]	0.31 [3.50]	0.18 [2.38]
Anomaly 1	53.61 [14.10]						33.78 [7.15]
Anomaly 2		44.95 [14.60]					30.90 [8.15]
Anomaly 3			21.74 [5.46]				3.63 [0.97]
Anomaly 4				23.66 [5.69]			1.96 [0.48]
Anomaly 5					11.69 [2.27]		-0.16 [-0.03]
Anomaly 6						15.30 [3.24]	-11.85 [-2.65]
mkt	-0.28 [-0.15]	-2.62 [-1.44]	-2.24 [-1.10]	-2.13 [-1.05]	-1.96 [-0.95]	-1.71 [-0.83]	-1.70 [-0.98]
smb	3.81 [1.42]	0.33 [0.12]	-3.51 [-1.19]	2.29 [0.75]	-3.58 [-1.19]	-2.20 [-0.74]	2.82 [1.05]
hml	-6.07 [-1.71]	-4.76 [-1.35]	-4.86 [-1.23]	-11.38 [-2.69]	-2.86 [-0.72]	-4.67 [-1.15]	-4.63 [-1.26]
rmw	-7.70 [-2.11]	-11.54 [-3.23]	-16.99 [-4.28]	-14.10 [-3.53]	-17.03 [-4.22]	-16.34 [-4.08]	-7.30 [-2.12]
cma	35.38 [6.12]	33.59 [5.83]	53.80 [8.24]	57.51 [9.28]	56.38 [6.59]	55.52 [7.74]	27.71 [3.60]
umd	5.51 [3.00]	3.91 [2.12]	8.73 [4.33]	9.76 [4.83]	9.13 [4.44]	8.11 [3.95]	4.28 [2.40]
# months	732	732	732	732	732	713	713
$\bar{R}^2(\%)$	47	47	35	35	32	33	52

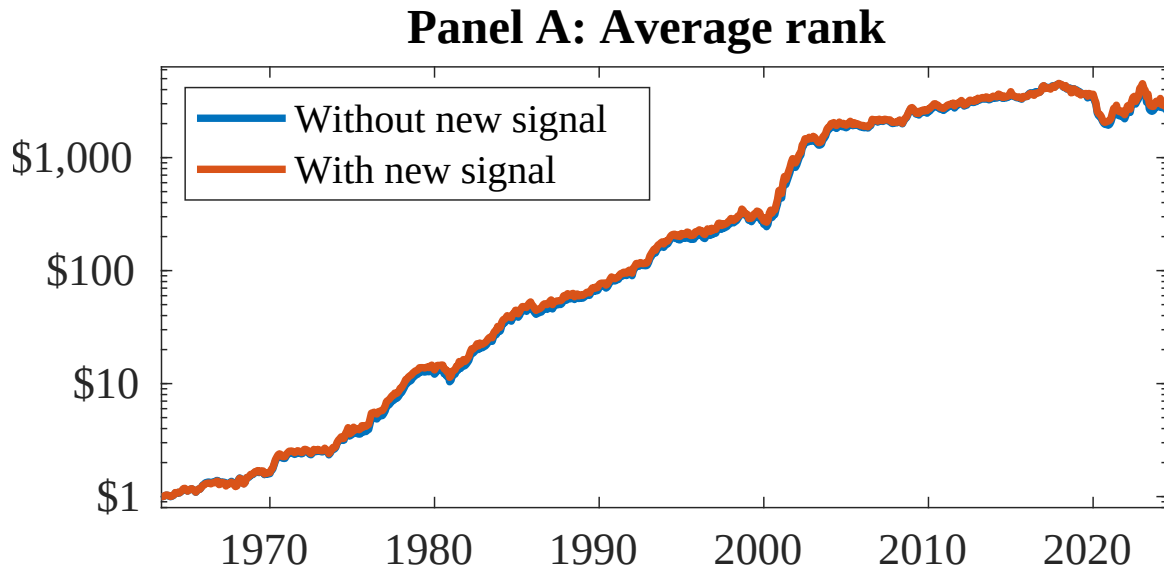


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as IOE. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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